

▼ LSTM Model

```
# LSTM (LONG SHORT-TERM MEMORY) FOR BEARING FAULT DETECTION

# INSTALL REQUIRED LIBRARIES FOR LSTM MODEL
!pip -q install tensorflow

import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
from sklearn.utils.class_weight import compute_class_weight
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout, Input, BatchNormalization, Bidirectional
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.regularizers import l2
from google.colab import files

# Upload all the required files ( train, val, test)

print("Upload your 3 CSV files:")
print("1. preprocessed_augmented_dataset_scaled.csv")
print("2. validation_set_scaled.csv")
print("3. test_set_scaled.csv")

uploaded = files.upload()

def find_file(keyword):
    for f in os.listdir():
        if keyword in f.lower():
            return f
    return None

train_file = find_file("augmented")
val_file = find_file("validation")
test_file = find_file("test")

print(f"Train: {train_file}")
print(f"Val: {val_file}")
print(f"Test: {test_file}")

# LOAD DATA

train_df = pd.read_csv(train_file)
val_df = pd.read_csv(val_file)
test_df = pd.read_csv(test_file)

X_train = train_df.iloc[:, :-1].values
y_train = train_df.iloc[:, -1].values

X_val = val_df.iloc[:, :-1].values
y_val = val_df.iloc[:, -1].values

X_test = test_df.iloc[:, :-1].values
y_test = test_df.iloc[:, -1].values

print(f"Training samples: {X_train.shape[0]}")
print(f"Validation samples: {X_val.shape[0]}")
print(f"Test samples: {X_test.shape[0]}")
print(f"Number of features: {X_train.shape[1]}")
print(f"Number of classes: {len(np.unique(y_train))}")

# =====
# PREPARE DATA
# =====

# One-hot encode labels
y_train_cat = to_categorical(y_train)
y_val_cat = to_categorical(y_val)
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y_test_cat = to_categorical(y_test)

# Reshape for LSTM: (samples, timesteps, features)
# Using 1 timestep with multiple features
X_train = X_train.reshape((X_train.shape[0], 1, X_train.shape[1]))
X_val = X_val.reshape((X_val.shape[0], 1, X_val.shape[1]))
X_test = X_test.reshape((X_test.shape[0], 1, X_test.shape[1]))

print(f"Input shape: {X_train.shape}")

# Class weights for imbalance
classes = np.unique(y_train)
class_weights = compute_class_weight('balanced', classes=classes, y=y_train)
class_weight_dict = dict(zip(classes, class_weights))

# BUILD LSTM MODEL

model = Sequential([
    Input(shape=(1, X_train.shape[2])),

    # First LSTM layer
    LSTM(64, return_sequences=True, activation='tanh', kernel_regularizer=l2(1e-4)),
    BatchNormalization(),
    Dropout(0.3),

    # Second LSTM layer
    LSTM(128, return_sequences=False, activation='tanh', kernel_regularizer=l2(1e-4)),
    BatchNormalization(),
    Dropout(0.3),

    # Dense layers
    Dense(128, activation='relu', kernel_regularizer=l2(1e-4)),
    Dropout(0.3),

    Dense(64, activation='relu', kernel_regularizer=l2(1e-4)),
    Dropout(0.3),

    Dense(y_train_cat.shape[1], activation='softmax')
])

model.compile(
    optimizer=Adam(learning_rate=0.001),
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

model.summary()

# TRAIN MODEL using the training dataset

early_stop = EarlyStopping(patience=15, restore_best_weights=True)
reduce_lr = ReduceLR0nPlateau(factor=0.5, patience=5, min_lr=1e-6)

history = model.fit(
    X_train, y_train_cat,
    epochs=80,
    batch_size=128,
    validation_data=(X_val, y_val_cat),
    callbacks=[early_stop, reduce_lr],
    class_weight=class_weight_dict,
    verbose=1
)

actual_epochs = len(history.history['accuracy'])
print(f"\nTrained for {actual_epochs}/{80} epochs")

# EVALUATION the model

y_pred = model.predict(X_test)
y_pred_classes = np.argmax(y_pred, axis=1)

accuracy = accuracy_score(y_test, y_pred_classes)
precision = precision_score(y_test, y_pred_classes, average='weighted')
recall = recall_score(y_test, y_pred_classes, average='weighted')
f1 = f1_score(y_test, y_pred_classes, average='weighted')

print("\n" + "="*70)
print("LSTM PERFORMANCE")
print("="*70)

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print(f"Accuracy:  {accuracy:.4f} ({accuracy*100:.2f}%)")
print(f"Precision: {precision:.4f}")
print(f"Recall:    {recall:.4f}")
print(f"F1 Score:  {f1:.4f}")
print("=="*70)

# VISUALIZATIONS

plt.figure(figsize=(15, 5))

# Accuracy Curves
plt.subplot(1, 3, 1)
plt.plot(history.history['accuracy'], label='Training Accuracy', linewidth=2)
plt.plot(history.history['val_accuracy'], label='Validation Accuracy', linewidth=2)
plt.title('LSTM Accuracy Curves', fontsize=14, fontweight='bold')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True, alpha=0.3)
plt.ylim([0, 1])

# Loss Curves
plt.subplot(1, 3, 2)
plt.plot(history.history['loss'], label='Training Loss', linewidth=2)
plt.plot(history.history['val_loss'], label='Validation Loss', linewidth=2)
plt.title('LSTM Loss Curves', fontsize=14, fontweight='bold')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.grid(True, alpha=0.3)

# Confusion Matrix
plt.subplot(1, 3, 3)
cm = confusion_matrix(y_test, y_pred_classes)
class_names = [str(i) for i in range(len(np.unique(y_test)))]
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=class_names, yticklabels=class_names)
plt.title('LSTM Confusion Matrix', fontsize=14, fontweight='bold')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')

plt.tight_layout()
plt.show()

# SAVE MODEL

model.save("lstm_model.keras")
print(f"\nModel saved to: lstm_model.keras")
```

Upload your 3 CSV files:
 1. preprocessed_augmented_dataset_scaled.csv
 2. validation_set_scaled.csv
 3. test_set_scaled.csv

No file chosen

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Saving test_set_scaled.csv to test_set_scaled.csv
 Saving validation_set_scaled.csv to validation_set_scaled.csv
 Saving preprocessed_augmented_dataset_scaled.csv to preprocessed_augmented_dataset_scaled.csv
 Train: preprocessed_augmented_dataset_scaled.csv
 Val: validation_set_scaled.csv
 Test: test_set_scaled.csv
 Training samples: 100000
 Validation samples: 320
 Test samples: 330
 Number of features: 19
 Number of classes: 10
 Input shape: (100000, 1, 19)
 Model: "sequential_1"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 1, 64)	21,504
batch_normalization (BatchNormalization)	(None, 1, 64)	256
dropout (Dropout)	(None, 1, 64)	0
lstm_1 (LSTM)	(None, 128)	98,816
batch_normalization_1 (BatchNormalization)	(None, 128)	512
dropout_1 (Dropout)	(None, 128)	0
dense (Dense)	(None, 128)	16,512
dropout_2 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 64)	8,256
dropout_3 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 10)	650

Total params: 146,506 (572.29 KB)

Trainable params: 146,122 (570.79 KB)

Non-trainable params: 384 (1.50 KB)

Epoch 1/80

782/782 ————— 18s 13ms/step - accuracy: 0.9040 - loss: 0.3036 - val_accuracy: 0.9500 - val_loss:

Epoch 2/80

782/782 ————— 10s 12ms/step - accuracy: 0.9570 - loss: 0.1612 - val_accuracy: 0.9594 - val_loss:

Epoch 3/80

782/782 ————— 10s 12ms/step - accuracy: 0.9654 - loss: 0.1360 - val_accuracy: 0.9563 - val_loss:

Epoch 4/80

782/782 ————— 8s 11ms/step - accuracy: 0.9701 - loss: 0.1178 - val_accuracy: 0.9531 - val_loss: (

Epoch 5/80

782/782 ————— 9s 12ms/step - accuracy: 0.9735 - loss: 0.1073 - val_accuracy: 0.9563 - val_loss: (

Epoch 6/80

782/782 ————— 10s 12ms/step - accuracy: 0.9747 - loss: 0.1011 - val_accuracy: 0.9531 - val_loss:

Epoch 7/80

782/782 ————— 10s 12ms/step - accuracy: 0.9774 - loss: 0.0924 - val_accuracy: 0.9594 - val_loss:

Epoch 8/80

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# =====
# ABLATION STUDY: LSTM FOR BEARING FAULT DETECTION
# Variants tested:
# 0. Full model (baseline)
# 1. No BatchNormalization
# 2. No Dropout
# 3. No L2 regularization
# 4. Single LSTM layer instead of stacked 2-layer
# 5. Bidirectional LSTM instead of unidirectional
# 6. No class weighting (imbalance handling removed)
# 7. Smaller LSTM units (64->32, 128->64)
# 8. No extra Dense head (drop post-LSTM dense layers)
# 9. Fixed higher learning rate, no ReduceLRonPlateau
# =====
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```
# !pip -q install tensorflow
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```
import os
import json
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf
```

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from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.utils.class_weight import compute_class_weight
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout, Input, BatchNormalization, Bidirectional
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.regularizers import l2

from google.colab import files
uploaded = files.upload()

SEED = 42
np.random.seed(SEED)
tf.random.set_seed(SEED)

# =====
# LOAD DATA
# =====

def find_file(keyword):
    for f in os.listdir():
        if keyword in f.lower():
            return f
    return None

train_file = find_file("augmented")
val_file = find_file("validation")
test_file = find_file("test")

print(f"Train: {train_file}")
print(f"Val: {val_file}")
print(f"Test: {test_file}")

train_df = pd.read_csv(train_file)
val_df = pd.read_csv(val_file)
test_df = pd.read_csv(test_file)

X_train_raw = train_df.iloc[:, :-1].values
y_train = train_df.iloc[:, -1].values

X_val_raw = val_df.iloc[:, :-1].values
y_val = val_df.iloc[:, -1].values

X_test_raw = test_df.iloc[:, :-1].values
y_test = test_df.iloc[:, -1].values

num_classes = len(np.unique(y_train))
y_train_cat = to_categorical(y_train, num_classes=num_classes)
y_val_cat = to_categorical(y_val, num_classes=num_classes)
y_test_cat = to_categorical(y_test, num_classes=num_classes)

# Reshape for LSTM: (samples, timesteps, features)
X_train = X_train_raw.reshape((X_train_raw.shape[0], 1, X_train_raw.shape[1]))
X_val = X_val_raw.reshape((X_val_raw.shape[0], 1, X_val_raw.shape[1]))
X_test = X_test_raw.reshape((X_test_raw.shape[0], 1, X_test_raw.shape[1]))

n_features = X_train.shape[2]

# Class weights for imbalance (used unless the variant explicitly disables it)
classes = np.unique(y_train)
class_weights = compute_class_weight('balanced', classes=classes, y=y_train)
class_weight_dict = dict(zip(classes, class_weights))

print(f"Train samples: {X_train.shape[0]} | Val: {X_val.shape[0]} | Test: {X_test.shape[0]}")
print(f"Features: {n_features} | Classes: {num_classes}")

# =====
# CONFIGURABLE MODEL BUILDER
# =====

DEFAULT_CONFIG = {
    "name": "baseline_full_model",
    "use_batchnorm": True,
    "use_dropout": True,
    "dropout_rate": 0.3,
    "use_l2": True,
    "l2_value": 1e-4,
    "num_lstm_layers": 2,          # 1 or 2
    "lstm_units": [64, 128],      # units per LSTM layer (len must match num_lstm_layers)
    "bidirectional": False,

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"use_dense_head": True,          # the two Dense(128)/Dense(64) layers after LSTM
"dense_units": [128, 64],
"use_class_weights": True,
"learning_rate": 0.001,
"use_lr_schedule": True,        # ReduceLR0nPlateau
}

def reg(cfg):
    return l2(cfg["l2_value"]) if cfg["use_l2"] else None

def build_model(cfg):
    layers = [Input(shape=(1, n_features))]

    for i in range(cfg["num_lstm_layers"]):
        units = cfg["lstm_units"][i]
        return_seq = (i < cfg["num_lstm_layers"] - 1) # only last LSTM returns False
        lstm_layer = LSTM(
            units,
            return_sequences=return_seq,
            activation='tanh',
            kernel_regularizer=reg(cfg),
        )
        if cfg["bidirectional"]:
            lstm_layer = Bidirectional(lstm_layer)
        layers.append(lstm_layer)

        if cfg["use_batchnorm"]:
            layers.append(BatchNormalization())
        if cfg["use_dropout"]:
            layers.append(Dropout(cfg["dropout_rate"]))

    if cfg["use_dense_head"]:
        for units in cfg["dense_units"]:
            layers.append(Dense(units, activation='relu', kernel_regularizer=reg(cfg)))
            if cfg["use_dropout"]:
                layers.append(Dropout(cfg["dropout_rate"]))

    layers.append(Dense(num_classes, activation='softmax'))

    model = Sequential(layers)
    model.compile(
        optimizer=Adam(learning_rate=cfg["learning_rate"]),
        loss='categorical_crossentropy',
        metrics=['accuracy'],
    )
    return model

def get_callbacks(cfg):
    callbacks = [EarlyStopping(patience=10, restore_best_weights=True, monitor='val_loss')]
    if cfg["use_lr_schedule"]:
        callbacks.append(ReduceLR0nPlateau(factor=0.5, patience=5, min_lr=1e-6))
    return callbacks

def run_experiment(cfg, epochs=40, batch_size=128, verbose=0):
    tf.keras.backend.clear_session()
    tf.random.set_seed(SEED)

    model = build_model(cfg)
    cw = class_weight_dict if cfg["use_class_weights"] else None

    history = model.fit(
        X_train, y_train_cat,
        epochs=epochs,
        batch_size=batch_size,
        validation_data=(X_val, y_val_cat),
        callbacks=get_callbacks(cfg),
        class_weight=cw,
        verbose=verbose,
    )

    y_pred = model.predict(X_test, verbose=0)
    y_pred_classes = np.argmax(y_pred, axis=1)

    metrics = {
        "variant": cfg["name"],
        "accuracy": accuracy_score(y_test, y_pred_classes),
        "precision": precision_score(y_test, y_pred_classes, average='weighted', zero_division=0),
        "recall": recall_score(y_test, y_pred_classes, average='weighted', zero_division=0),
    }

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        "f1_score": f1_score(y_test, y_pred_classes, average='weighted', zero_division=0),
        "epochs_run": len(history.history['accuracy']),
        "final_val_accuracy": history.history['val_accuracy'][-1],
        "final_val_loss": history.history['val_loss'][-1],
        "trainable_params": model.count_params(),
    }
    return metrics, history

# =====
# DEFINE ABLATION VARIANTS
# =====

def make_cfg(overrides):
    cfg = dict(DEFAULT_CONFIG)
    cfg.update(overrides)
    return cfg

ablation_variants = [
    make_cfg({"name": "0_baseline_full_model"}),

    make_cfg({"name": "1_no_batchnorm", "use_batchnorm": False}),

    make_cfg({"name": "2_no_dropout", "use_dropout": False}),

    make_cfg({"name": "3_no_l2_regularization", "use_l2": False}),

    make_cfg({
        "name": "4_single_lstm_layer",
        "num_lstm_layers": 1,
        "lstm_units": [128],
    }),

    make_cfg({"name": "5_bidirectional_lstm", "bidirectional": True}),

    make_cfg({"name": "6_no_class_weights", "use_class_weights": False}),

    make_cfg({
        "name": "7_smaller_lstm_units",
        "lstm_units": [32, 64],
    }),

    make_cfg({"name": "8_no_dense_head", "use_dense_head": False}),

    make_cfg({
        "name": "9_fixed_higher_lr_no_schedule",
        "learning_rate": 0.005,
        "use_lr_schedule": False,
    }),
]

# =====
# RUN ALL EXPERIMENTS
# =====

results = []
histories = {}

for cfg in ablation_variants:
    print("\n" + "=" * 70)
    print(f"Running variant: {cfg['name']}")
    print("=" * 70)
    metrics, history = run_experiment(cfg, epochs=40, batch_size=128, verbose=1)
    results.append(metrics)
    histories[cfg["name"]] = history.history
    print(f"    -> Accuracy: {metrics['accuracy']:.4f} | F1: {metrics['f1_score']:.4f} "
          f"| Epochs: {metrics['epochs_run']} | Params: {metrics['trainable_params']:,}")

results_df = pd.DataFrame(results).sort_values("f1_score", ascending=False).reset_index(drop=True)

print("\n" + "=" * 70)
print("ABLATION STUDY RESULTS (sorted by F1 score)")
print("=" * 70)
print(results_df.to_string(index=False))

results_df.to_csv("ablation_results.csv", index=False)
with open("ablation_histories.json", "w") as f:
    json.dump(histories, f)

print("\nSaved: ablation_results.csv, ablation_histories.json")

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# =====
# VISUALIZATIONS
# =====

baseline_row = results_df[results_df["variant"] == "0_baseline_full_model"].iloc[0]

# --- Bar chart: metric comparison across variants ---
plt.figure(figsize=(14, 6))
metrics_to_plot = ["accuracy", "precision", "recall", "f1_score"]
x = np.arange(len(results_df))
width = 0.2

for i, metric in enumerate(metrics_to_plot):
    plt.bar(x + i * width, results_df[metric], width, label=metric)

plt.xticks(x + width * 1.5, results_df["variant"], rotation=45, ha='right')
plt.ylabel("Score")
plt.title("Ablation Study: Metric Comparison Across Variants", fontsize=14, fontweight='bold')
plt.legend()
plt.grid(True, axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig("ablation_metric_comparison.png", dpi=150)
plt.show()

# --- Delta-from-baseline chart (the key ablation insight) ---
plt.figure(figsize=(12, 6))
delta_f1 = results_df["f1_score"] - baseline_row["f1_score"]
colors = ['#d62728' if v < 0 else '#2ca02c' for v in delta_f1]
plt.barh(results_df["variant"], delta_f1, color=colors)
plt.axvline(0, color='black', linewidth=0.8)
plt.xlabel("Change in F1 Score vs. Baseline")
plt.title("Impact of Removing/Changing Each Component (relative to full model)",
          fontsize=13, fontweight='bold')
plt.grid(True, axis='x', alpha=0.3)
plt.tight_layout()
plt.savefig("ablation_delta_from_baseline.png", dpi=150)
plt.show()

# --- Training curves overlay for accuracy ---
plt.figure(figsize=(14, 6))
for name, hist in histories.items():
    plt.plot(hist['val_accuracy'], label=name, alpha=0.8)
plt.title("Validation Accuracy Curves – All Variants", fontsize=14, fontweight='bold')
plt.xlabel("Epoch")
plt.ylabel("Validation Accuracy")
plt.legend(fontsize=8, loc='lower right')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("ablation_val_accuracy_curves.png", dpi=150)
plt.show()

print("\nSaved plots: ablation_metric_comparison.png, ablation_delta_from_baseline.png, "
      "ablation_val_accuracy_curves.png")

# =====
# SUMMARY: WHICH COMPONENTS MATTER MOST
# =====
print("\n" + "=" * 70)
print("SUMMARY: Component impact (F1 drop when removed/changed, vs baseline)")
print("=" * 70)
summary = results_df.copy()
summary["f1_delta_vs_baseline"] = summary["f1_score"] - baseline_row["f1_score"]
summary = summary[summary["variant"] != "0_baseline_full_model"]
summary = summary.sort_values("f1_delta_vs_baseline")
print(summary[["variant", "f1_delta_vs_baseline", "trainable_params"]].to_string(index=False))

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Train: preprocessed_augmented_dataset_scaled.csv
Val: validation_set_scaled.csv
Test: test_set_scaled.csv
Train samples: 100000 | Val: 320 | Test: 330
Features: 19 | Classes: 10

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Running variant: 0_baseline_full_model

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Epoch 1/40
782/782 ----- 15s 13ms/step - accuracy: 0.9008 - loss: 0.3174 - val_accuracy: 0.9375 - val_loss: 0.
Epoch 2/40
782/782 ----- 10s 12ms/step - accuracy: 0.9565 - loss: 0.1629 - val_accuracy: 0.9531 - val_loss: 0.
Epoch 3/40
782/782 ----- 9s 11ms/step - accuracy: 0.9647 - loss: 0.1386 - val_accuracy: 0.9500 - val_loss: 0.
Epoch 4/40
782/782 ----- 10s 12ms/step - accuracy: 0.9694 - loss: 0.1240 - val_accuracy: 0.9563 - val_loss: 0.
Epoch 5/40
782/782 ----- 10s 12ms/step - accuracy: 0.9732 - loss: 0.1092 - val_accuracy: 0.9469 - val_loss: 0.
Epoch 6/40
782/782 ----- 10s 12ms/step - accuracy: 0.9756 - loss: 0.1014 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 7/40
782/782 ----- 9s 12ms/step - accuracy: 0.9774 - loss: 0.0946 - val_accuracy: 0.9438 - val_loss: 0.
Epoch 8/40
782/782 ----- 10s 12ms/step - accuracy: 0.9814 - loss: 0.0804 - val_accuracy: 0.9531 - val_loss: 0.
Epoch 9/40
782/782 ----- 10s 12ms/step - accuracy: 0.9819 - loss: 0.0763 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 10/40
782/782 ----- 11s 14ms/step - accuracy: 0.9824 - loss: 0.0742 - val_accuracy: 0.9563 - val_loss: 0.
Epoch 11/40
782/782 ----- 19s 12ms/step - accuracy: 0.9838 - loss: 0.0698 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 12/40
782/782 ----- 10s 12ms/step - accuracy: 0.9843 - loss: 0.0670 - val_accuracy: 0.9563 - val_loss: 0.
-> Accuracy: 0.9424 | F1: 0.9422 | Epochs: 12 | Params: 146,506

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Running variant: 1_no_batchnorm

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Epoch 1/40
782/782 ----- 11s 11ms/step - accuracy: 0.8693 - loss: 0.4201 - val_accuracy: 0.9531 - val_loss: 0.
Epoch 2/40
782/782 ----- 10s 10ms/step - accuracy: 0.9588 - loss: 0.1731 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 3/40
782/782 ----- 9s 9ms/step - accuracy: 0.9655 - loss: 0.1484 - val_accuracy: 0.9531 - val_loss: 0.2
Epoch 4/40
782/782 ----- 10s 9ms/step - accuracy: 0.9701 - loss: 0.1338 - val_accuracy: 0.9531 - val_loss: 0.
Epoch 5/40
782/782 ----- 8s 10ms/step - accuracy: 0.9725 - loss: 0.1251 - val_accuracy: 0.9625 - val_loss: 0.
Epoch 6/40
782/782 ----- 7s 9ms/step - accuracy: 0.9744 - loss: 0.1190 - val_accuracy: 0.9594 - val_loss: 0.2
Epoch 7/40
782/782 ----- 8s 10ms/step - accuracy: 0.9750 - loss: 0.1138 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 8/40
782/782 ----- 8s 10ms/step - accuracy: 0.9768 - loss: 0.1082 - val_accuracy: 0.9625 - val_loss: 0.
Epoch 9/40
782/782 ----- 7s 9ms/step - accuracy: 0.9768 - loss: 0.1054 - val_accuracy: 0.9469 - val_loss: 0.2
Epoch 10/40
782/782 ----- 8s 10ms/step - accuracy: 0.9777 - loss: 0.1034 - val_accuracy: 0.9500 - val_loss: 0.
Epoch 11/40
782/782 ----- 7s 10ms/step - accuracy: 0.9783 - loss: 0.1005 - val_accuracy: 0.9531 - val_loss: 0.
Epoch 12/40
782/782 ----- 8s 10ms/step - accuracy: 0.9782 - loss: 0.0982 - val_accuracy: 0.9531 - val_loss: 0.

```

Final LSTM Model

```

# =====
# FINAL LSTM MODEL - BEARING FAULT DETECTION
# =====

import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix
)
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout, Input, BatchNormalization
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
from tensorflow.keras.optimizers import Adam

tf.random.set_seed(42)

```

```

np.random.seed(42)

# =====
# 1. UPLOAD DATASET FILES
# =====

print("\nUpload your 3 CSV files:")
print("1. preprocessed_augmented_dataset_scaled.csv")
print("2. validation_set_scaled.csv")
print("3. test_set_scaled.csv")

from google.colab import files
uploaded = files.upload()

def find_file(keyword):
    for f in os.listdir():
        if keyword in f.lower():
            return f
    return None

train_file = find_file("augmented")
val_file = find_file("validation")
test_file = find_file("test")

missing = []
if train_file is None:
    missing.append("training set (file containing 'augmented')")
if val_file is None:
    missing.append("validation set (file containing 'validation')")
if test_file is None:
    missing.append("test set (file containing 'test')")

if missing:
    raise FileNotFoundError(
        "Missing required files: " + ", ".join(missing) +
        "\nCurrent directory contents: " + ", ".join(os.listdir())
    )

print(f"Train file: {train_file}")
print(f"Val file: {val_file}")
print(f"Test file: {test_file}")

# =====
# 2. LOAD AND PREPARE DATA
# =====

train_df = pd.read_csv(train_file)
val_df = pd.read_csv(val_file)
test_df = pd.read_csv(test_file)

X_train = train_df.iloc[:, :-1].values
y_train = train_df.iloc[:, -1].values

X_val = val_df.iloc[:, :-1].values
y_val = val_df.iloc[:, -1].values

X_test = test_df.iloc[:, :-1].values
y_test = test_df.iloc[:, -1].values

print(f"Training samples: {X_train.shape[0]}")
print(f"Validation samples: {X_val.shape[0]}")
print(f"Test samples: {X_test.shape[0]}")
print(f"Number of features: {X_train.shape[1]}")
print(f"Number of classes: {len(np.unique(y_train))}")

y_train_cat = to_categorical(y_train)
y_val_cat = to_categorical(y_val)
y_test_cat = to_categorical(y_test)

X_train = X_train.reshape((X_train.shape[0], 1, X_train.shape[1]))
X_val = X_val.reshape((X_val.shape[0], 1, X_val.shape[1]))
X_test = X_test.reshape((X_test.shape[0], 1, X_test.shape[1]))

print(f"Input shape: {X_train.shape}")

# =====
# 3. BUILD FINAL OPTIMIZED MODEL
# Ablation findings integrated below:
# - L2 regularization removed (no improvement, 95.15% vs 94.24%)
# - Class weights removed (hurt performance by 0.3%)
# - Dropout and BatchNorm kept (each adds ~0.6% accuracy)
# - Two LSTM layers kept (single layer loses 1.2% accuracy)
# - Dense(64) kept (loses 1.2% accuracy if removed)

```

```

# - Bidirectional rejected (overfits with 1-timestep input)
# =====

model = Sequential([
    Input(shape=(1, X_train.shape[2])),

    # LSTM(64) returning sequences for stacking (kept; single LSTM costs -1.2%)
    LSTM(64, return_sequences=True, activation='tanh'),
    BatchNormalization(), # Kept; removing costs -0.6% accuracy
    Dropout(0.3),         # Kept; removing costs -0.6% accuracy

    # Second LSTM layer (kept for hierarchical feature extraction)
    LSTM(128, return_sequences=False, activation='tanh'),
    BatchNormalization(),
    Dropout(0.3),

    # Dense layers (Dense64 kept; removing costs -1.2% accuracy)
    Dense(128, activation='relu'),
    Dropout(0.3),

    Dense(64, activation='relu'),
    Dropout(0.3),

    # Output layer (10 bearing fault classes)
    Dense(y_train_cat.shape[1], activation='softmax')
])

# L2 regularization omitted per ablation findings (no benefit)
model.compile(
    optimizer=Adam(learning_rate=0.001),
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

model.summary()

# =====
# 4. TRAIN MODEL
# =====

early_stop = EarlyStopping(patience=15, restore_best_weights=True)
reduce_lr = ReduceLROnPlateau(factor=0.5, patience=5, min_lr=1e-6)

history = model.fit(
    X_train, y_train_cat,
    epochs=80,
    batch_size=128,
    validation_data=(X_val, y_val_cat),
    callbacks=[early_stop, reduce_lr],
    # Class weights removed per ablation (tied best at 95.15% without them)
    verbose=1
)

actual_epochs = len(history.history['accuracy'])
print(f"\nTrained for {actual_epochs} epochs")

# =====
# 5. EVALUATE ON TEST SET
# =====

y_pred_probs = model.predict(X_test)
y_pred_classes = np.argmax(y_pred_probs, axis=1)

accuracy = accuracy_score(y_test, y_pred_classes)
precision = precision_score(y_test, y_pred_classes, average='weighted')
recall = recall_score(y_test, y_pred_classes, average='weighted')
f1 = f1_score(y_test, y_pred_classes, average='weighted')

print("\n" + "=" * 80)
print("FINAL MODEL PERFORMANCE")
print("=" * 80)
print(f"Accuracy:  {accuracy:.4f} ({accuracy*100:.2f}%)")
print(f"Precision: {precision:.4f}")
print(f"Recall:    {recall:.4f}")
print(f"F1 Score:  {f1:.4f}")
print("=" * 80)

print("\nAblation benchmark comparison:")
print(f" Baseline (with L2 + class weights) : 94.24%")
print(f" Optimized (without L2 + class weights): {accuracy*100:.2f}%")
print(f" Improvement: +{(accuracy - 0.9424)*100:.2f}%")

# =====

```

```
# 6. VISUALIZATION
# =====

plt.figure(figsize=(18, 5))

plt.subplot(1, 3, 1)
plt.plot(history.history['accuracy'], label='Training Accuracy', linewidth=2)
plt.plot(history.history['val_accuracy'], label='Validation Accuracy', linewidth=2)
plt.title('Accuracy Curves', fontsize=14, fontweight='bold')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True, alpha=0.3)
plt.ylim([0.85, 1.0])

plt.subplot(1, 3, 2)
plt.plot(history.history['loss'], label='Training Loss', linewidth=2)
plt.plot(history.history['val_loss'], label='Validation Loss', linewidth=2)
plt.title('Loss Curves', fontsize=14, fontweight='bold')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.grid(True, alpha=0.3)

plt.subplot(1, 3, 3)
cm = confusion_matrix(y_test, y_pred_classes)
class_names = [str(i) for i in range(len(np.unique(y_test)))]
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=class_names, yticklabels=class_names)
plt.title('Confusion Matrix', fontsize=14, fontweight='bold')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')

plt.tight_layout()
plt.savefig("final_lstm_evaluation.png", dpi=150)
plt.show()

# =====
# 7. SAVE MODEL
# =====

model.save("lstm_model_final.keras")
print("\nModel saved as: lstm_model_final.keras")
print("Evaluation plot saved as: final_lstm_evaluation.png")

782/782 ————— 9s 11ms/step - accuracy: 0.9885 - loss: 0.0457 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 13/40
782/782 ————— 8s 10ms/step - accuracy: 0.9890 - loss: 0.0445 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 14/40
782/782 ————— 9s 11ms/step - accuracy: 0.9897 - loss: 0.0430 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 15/40
782/782 ————— 8s 11ms/step - accuracy: 0.9899 - loss: 0.0420 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 16/40
782/782 ————— 8s 10ms/step - accuracy: 0.9909 - loss: 0.0401 - val_accuracy: 0.9625 - val_loss: 0.
Epoch 17/40
782/782 ————— 9s 11ms/step - accuracy: 0.9907 - loss: 0.0407 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 18/40
782/782 ————— 10s 11ms/step - accuracy: 0.9923 - loss: 0.0368 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 19/40
782/782 ————— 8s 10ms/step - accuracy: 0.9928 - loss: 0.0349 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 20/40
782/782 ————— 9s 11ms/step - accuracy: 0.9927 - loss: 0.0349 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 21/40
782/782 ————— 10s 11ms/step - accuracy: 0.9930 - loss: 0.0343 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 22/40
782/782 ————— 8s 10ms/step - accuracy: 0.9934 - loss: 0.0333 - val_accuracy: 0.9594 - val_loss: 0.
-> Accuracy: 0.9455 | F1: 0.9458 | Epochs: 22 | Params: 122,378

=====
Running variant: 9_fixed_higher_lr_no_schedule
=====

Epoch 1/40
782/782 ————— 14s 12ms/step - accuracy: 0.9355 - loss: 0.2402 - val_accuracy: 0.9469 - val_loss: 0.
Epoch 2/40
782/782 ————— 10s 13ms/step - accuracy: 0.9641 - loss: 0.1636 - val_accuracy: 0.9344 - val_loss: 0.
Epoch 3/40
782/782 ————— 20s 13ms/step - accuracy: 0.9692 - loss: 0.1541 - val_accuracy: 0.9531 - val_loss: 0.
Epoch 4/40
782/782 ————— 10s 13ms/step - accuracy: 0.9709 - loss: 0.1490 - val_accuracy: 0.9375 - val_loss: 0.
Epoch 5/40
782/782 ————— 9s 12ms/step - accuracy: 0.9728 - loss: 0.1484 - val_accuracy: 0.9563 - val_loss: 0.
Epoch 6/40
782/782 ————— 10s 13ms/step - accuracy: 0.9735 - loss: 0.1512 - val_accuracy: 0.9594 - val_loss: 0.
Epoch 7/40
782/782 ————— 11s 13ms/step - accuracy: 0.9746 - loss: 0.1482 - val_accuracy: 0.9531 - val_loss: 0.
Epoch 8/40
782/782 ————— 10s 11ms/step - accuracy: 0.9765 - loss: 0.1446 - val_accuracy: 0.9594 - val_loss: 0.
```

```

Epoch 9/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 10/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 11/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 12/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 13/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 14/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 15/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 16/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 17/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 18/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 19/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 20/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 21/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 22/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 23/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 24/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 25/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 26/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 27/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 28/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 29/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 30/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 31/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 32/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 33/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 34/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 35/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 36/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 37/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 38/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 39/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441
Epoch 40/40
0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441

```

Choose Files

No file chosen

0.12s/step - accuracy: 0.9750 - loss: 0.1517 - val_accuracy: 0.9500 - val_loss: 0.1441